New Calculations of Numerical Invariants Related to the Big Finitistic Dimension

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Flash Talk

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Introduction

Throughout, let k be a(n algebraically complete) field and A be a finite-dimensional k-algebra.

Conjecture

The **big finitistic dimension conjecture (big FDC)** claims that the **big finitistic dimension** of A, denoted by $\mathrm{FinDim}(A)$:

$$\mathrm{FinDim}(A) := \sup\{\mathrm{pd}(M) : M \in \mathrm{Mod}\, A \text{ such that } \mathrm{pd}(M) < \infty\}$$

is finite for all A.

A Good Paper

THE FINITISTIC DIMENSION CONJECTURES – A TALE OF 3.5 DECADES

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Key words: Finite dimensional algebra, finitistic dimension, projective resolution, projective dimension, syzygy, contravariant finiteness.

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A. Facchini and C. Menini (eds.), Abelian Groups and Modules, 501–517.
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New Invariants

Given (right) A-module M, denote $\Omega^n(M)$ as the nth syzygy M.

Definition (Gélinas, Guo and Igusa)

The k-**delooping level** of M is the least integer n, if exists, such that there exists module N with $\Omega^n(M)$ as a summand of $\Omega^{n+k}(N)$. We write $k\text{-dell}(M) = n$.

Definition (Guo and Igusa)

The **derived delooping level** of an A-module M is:

$$\operatorname{ddell} M = \inf\{m \in \mathbb{N} \mid \exists n \leq m \text{ and an exact sequence in mod } A$$
$$0 \to C_n \to C_{n-1} \to \cdots \to C_1 \to C_0 \to M \to 0$$
$$\text{where } (i+1)\text{-dell}\, C_i \leq m-i, i = 0,1,\ldots,n\}$$

Results on Invariants

Definition

The k-delooping level of an *algebra* is the supremum of delooping level over all simple modules.

Theorem (Gélinas, Guo and Igusa)

$\mathrm{FinDim}(A^{op}) \leq \mathrm{ddell}(A) \leq k\text{-}\mathrm{dell}(A)$ *for all* k.

New Calculations

Barrios, Lanzilotta, Mata considered a collection of quiver algebras depending on two positive integer $k < n$.

Theorem (L.)

For this collection of algebras, we have $1\text{-dell}(A) = O(n)$ *and* $\text{ddell}(A) = \text{FinDim}(A^{op}) = O(k)$.

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$O(n)$ just means $\left\lfloor \frac{n+1}{2} \right\rfloor$ or something like that, I am lazy.

Remarks

- The algebra is monomial. So FinDim can be calculated with high accuracy due to a result of Husigen-Zimmermann, and we can argue that they are equal.
- Finding an example where $\mathrm{ddell}(A) \neq \mathrm{FinDim}(A^{op})$ is open.

Thank you!